WORKED SOLUTION

STEP III 2003, Question 1

Pure • Inverse Hyperbolic Integration

STEP III 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q1

The key idea

The first two results are careful chain-rule calculations. The useful thing they produce is a template: the derivative of each inverse function has the shape $\dfrac{\text{constant}}{(\text{linear})\sqrt{\text{linear}}}$, and that is exactly the shape of both integrands. So for each integral we choose a and b to match the template. The bracket outside the root tells us which formula to use. If it is $x+b$, the larger constant, we use arcsin; if it is $x+a$, the smaller one, we use arcosh.

Solution

The two derivatives

Let $y = \dfrac{x+a}{x+b}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+b)-(x+a)}{(x+b)^2} = \frac{b-a}{(x+b)^2}.$$

Since $x+a>0$ and $x+b>0$ we have $y>0$, and since $b>a$ the numerator is smaller than the denominator, so $0<y<1$ and $\arcsin y$ is differentiable there. Next,

$$1-y^2 = \frac{(x+b)^2-(x+a)^2}{(x+b)^2} = \frac{(b-a)(2x+a+b)}{(x+b)^2},$$

using the difference of two squares. Since $x+b>0$, the square root of $(x+b)^2$ is $x+b$ itself, so

$$\sqrt{1-y^2} = \frac{\sqrt{b-a}\,\sqrt{a+b+2x}}{x+b}.$$

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\arcsin\left(\frac{x+a}{x+b}\right) = \frac{1}{\sqrt{1-y^2}}\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+b}{\sqrt{b-a}\,\sqrt{a+b+2x}}\cdot\frac{b-a}{(x+b)^2}$$
$$= \frac{\sqrt{b-a}}{(x+b)\sqrt{a+b+2x}},$$

since $(b-a)/\sqrt{b-a} = \sqrt{b-a}$. This is the required result.

For the arcosh, let $z = \dfrac{x+b}{x+a}$. Here $z > 1$, because $x+b > x+a > 0$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{(x+a)-(x+b)}{(x+a)^2} = -\frac{b-a}{(x+a)^2}, \qquad z^2-1 = \frac{(b-a)(2x+a+b)}{(x+a)^2}.$$

Since $x+a>0$, $\sqrt{z^2-1} = \sqrt{b-a}\,\sqrt{a+b+2x}\,/(x+a)$, and so

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh}\left(\frac{x+b}{x+a}\right) = \frac{x+a}{\sqrt{b-a}\,\sqrt{a+b+2x}}\cdot\left(-\frac{b-a}{(x+a)^2}\right) = -\frac{\sqrt{b-a}}{(x+a)\sqrt{a+b+2x}}.$$

The two results look almost alike. The differences are the sign and which bracket sits outside the root.

Ask yourself Which of the hypotheses did the arcsin calculation actually use?

We needed $x+b>0$ to simplify $\sqrt{(x+b)^2}$, and we needed $1-y^2>0$. From the factorised form, $1-y^2>0$ holds exactly when $a+b+2x>0$, given $b>a$. We never used $x+a>0$ itself. So the arcsin formula holds whenever $x+b>0$ and $a+b+2x>0$, even if $x+a$ is negative, in which case y lies in $(-1,0]$. This matters in (ii).

Part (i)

The bracket outside the root is $x+1$, so we try to match it with $x+a$ in the arcosh formula. Write $\sqrt{x+3} = \sqrt{2x+6}\,/\sqrt{2}$, so we need $a+b=6$ as well as $a=1$. That gives $b=5$, so $b-a=4$. With these values the hypotheses $x+a>0$, $x+b>0$ and $b>a$ all hold for $x>-1$, and

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh}\left(\frac{x+5}{x+1}\right) = -\frac{2}{(x+1)\sqrt{2x+6}} = -\frac{\sqrt{2}}{(x+1)\sqrt{x+3}}.$$

Dividing by $-\sqrt{2}$,

$$\int \frac{1}{(x+1)\sqrt{x+3}}\,\mathrm{d}x = -\frac{1}{\sqrt{2}}\operatorname{arcosh}\left(\frac{x+5}{x+1}\right) + C.$$

Part (ii)

Now the bracket outside is $x+3$, so we match it with $x+b$ in the arcsin formula: $b=3$. Inside the root we need $a+b+2x = 2(x+1)$, so $a+b=2$ and $a=-1$. Then $b-a=4$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}\arcsin\left(\frac{x-1}{x+3}\right)=\frac{2}{(x+3)\sqrt{2x+2}}=\frac{\sqrt{2}}{(x+3)\sqrt{x+1}}.$$

Ask yourself With $a=-1$, the hypothesis $x+a>0$ requires $x>1$. Does our answer still hold on the whole range $x>-1$?

Yes, it does. As we saw above, the arcsin formula needs only $x+b>0$ and $a+b+2x>0$. Here these say $x+3>0$ and $2x+2>0$, and both hold for every $x>-1$. On this range $(x-1)/(x+3)$ goes from -1 up towards 1, so the arcsin is defined. Hence, for $x>-1$,

$$\int\frac{1}{(x+3)\sqrt{x+1}}\,\mathrm{d}x=\frac{1}{\sqrt{2}}\arcsin\left(\frac{x-1}{x+3}\right)+C.$$

Otherwise: a substitution

Both integrals also give way to $u=\sqrt{x+c}$, where $x+c$ is the expression under the root, since then $\mathrm{d}x=2u\,\mathrm{d}u$. In (ii), with $u=\sqrt{x+1}$ the integral becomes $\int\frac{2}{u^2+2}\,\mathrm{d}u=\sqrt{2}\arctan(\sqrt{(x+1)/2})+C$. This differs from our answer only by the constant $-\pi/(2\sqrt{2})$. In (i), with $u=\sqrt{x+3}$ we get $\int\frac{2}{u^2-2}\,\mathrm{d}u=\frac{1}{\sqrt{2}}\ln\left|\frac{\sqrt{x+3}-\sqrt{2}}{\sqrt{x+3}+\sqrt{2}}\right|+C$. This is exactly the arcosh answer written in logarithms.

Checks

- Differentiate the answers back. For example, at $x=0$ the derivative of $\frac{1}{\sqrt{2}}\arcsin\frac{x-1}{x+3}$ should be $\frac{1}{3}$, and the formula above gives $\frac{\sqrt{2}}{\sqrt{2}\cdot3\cdot1}=\frac{1}{3}$.
- Numerically, $\int_0^1\frac{\mathrm{d}x}{(x+1)\sqrt{x+3}}\approx0.374543$ and $\int_0^1\frac{\mathrm{d}x}{(x+3)\sqrt{x+1}}\approx0.240301$, by Simpson's rule. Both values agree with the antiderivatives to better than 10^{-12}. The agreement also holds on intervals reaching down to $x=-0.9$, inside the range $-1<x<1$ where $x+a<0$ in (ii).
- Signs: the integrand in (i) is positive, so its integral should be increasing. Our answer is, because $\frac{x+5}{x+1}$ decreases as x increases, so minus its arcosh increases.

Common slips

- Matching the wrong bracket to the wrong formula: in (i) the $x+1$ outside the root is the *smaller* constant, so it has to be $x+a$, which means arcosh, not arcsin.
- Forgetting the factor $\sqrt{2}$ that comes from writing $\sqrt{x+3}$ as $\sqrt{2x+6}/\sqrt{2}$.
- Dropping the minus sign in the arcosh derivative, which comes from $\mathrm{d}z/\mathrm{d}x<0$.

Beyond STEP

The two answers to (ii), one with arcsin and one with arctan, differ by a constant. That is guaranteed because two functions with the same derivative on an *interval* differ by a constant, a consequence of the mean value theorem. On a set with two separate pieces the constants could be different. The logarithm in (i) is no accident either: inverse hyperbolic functions are logarithms in disguise, and that is why integrals of $1/\sqrt{\text{quadratic}}$ give either inverse trigonometric functions or logarithms. Abbott's *Understanding Analysis* proves the mean value theorem carefully, and Siklos's *Advanced Problems in Mathematics* has more integrals of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.